State, Motion, Boundary: A Mechanistic Account of Convergence and Failure in Transformer Multiple-Choice Decisions

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Abstract

We introduce three quantities—state, motion, and boundary—that fully describe the layer-by-layer trajectory of a transformer's answer on multiple-choice benchmarks. The state is the logit margin between the model's preferred answer and its strongest competitor at each layer. Motion is the per-layer change in that margin. Boundary is its absolute magnitude, measuring proximity to a decision flip. Across three 7–8B instruction-tuned transformers, these primitives separate prompts into four trajectory types—stable-correct, stable-wrong, unstable-correct, and unstable-wrong—producing a phase diagram that replicates across models and knowledge domains. We decompose motion into attention-driven routing and MLP-driven injection, achieving reconstruction $R^2 \geq 0.70$ on held-out data. Causal validation via component ablation, activation patching, and span-deletion experiments—controlled against shuffled and sign-flipped baselines with Benjamini–Hochberg correction—confirms directionally consistent, finite effects on the decision margin.

1. Introduction

A transformer answering a multiple-choice question does not make a single decision. It makes one decision per layer. At layer zero the output logits are largely uninformed; by the final layer they commit to an answer. What happens in between—whether the trajectory converges smoothly, oscillates, or drifts into the wrong basin—determines whether the model answers correctly.

Prior work in mechanistic interpretability has illuminated transformer internals through probing classifiers (Belinkov, 2022), causal tracing (Meng et al., 2022), and circuit-level analysis (Olsson et al., 2022; Conmy et al., 2023). These approaches identify what a model knows and where it stores that knowledge, yet typically treat the final output as a binary event rather than examining the dynamic process by which the model arrives at it across its full depth.

We propose a complementary perspective. Rather than asking what a model has learned, we ask: what does the trajectory of a model's decision look like as it propagates through layers, and can we predict convergence outcomes from a small set of trajectory-level quantities?

Our answer is yes. We define three primitives—state (the logit margin δ), motion (per-layer drift g), and boundary (the absolute margin $|\delta|$)—and show that they suffice to classify decision trajectories into four outcome types that replicate across models and domains. We then decompose motion into attention and MLP contributions and validate the resulting mechanistic claims causally.

Contributions.

1. A physics-inspired framework for analyzing transformer decisions as trajectories in logit-margin space, characterized by three measurable quantities.

2. A phase diagram with four well-separated trajectory types, replicated across three architectures and multiple knowledge domains.

3. A linear decomposition of per-layer drift into attention and MLP components with $R^2 \geq 0.70$ on held-out data and a mechanistic interpretation of routing versus injection.

4. Causal validation through ablation, patching, and span-deletion experiments with multiple-comparison-corrected tests and negative controls.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

2. The Three Primitives

We study four-choice (A/B/C/D) questions. At each transformer layer $l \in \{0, \ldots, L\}$, we extract logits for the four answer tokens and define all quantities in this subspace.

2.1. State: The Decision Margin

Let $z_c(l)$ denote the logit for the correct answer and $z_{k^*}(l)$ the logit for the strongest competitor—the highest-scoring incorrect option. The decision margin is:

$$\delta(l) = z_c(l) - z_{k^*}(l). \tag{1}$$

When $\delta(l) > 0$ the model favors the correct answer; the sign at the final layer determines correctness.

Design decision. We track competitor identity $k^*(l)$ per layer rather than fixing it globally. A sign flip in δ can arise from evidence reversal or competitor switch; we report both.

2.2. Motion: Layer-by-Layer Drift

The drift is the per-layer change in decision margin:

$$g(l) = \delta(l{+}1) - \delta(l), \tag{2}$$

with $g(L) \equiv 0$. Positive drift moves toward the correct answer.

2.3. Boundary Proximity

The boundary distance is the unsigned state:

$$b(l) = |\delta(l)|. \tag{3}$$

Small b means proximity to a decision flip.

2.4. Convergence and Trajectory Types

A trajectory is stable if, within a tail window of the last τ layers, $\delta(l)$ maintains a consistent sign and $b(l) \geq b_{\min}$ throughout. Combined with final-layer correctness this yields four types: stable-correct, stable-wrong, unstable-correct, unstable-wrong.

We use $\tau = 8$, require zero sign flips, and enforce $b_{\min} = 0.3$. These thresholds were pre-registered. Sensitivity analysis appears in Appendix A.

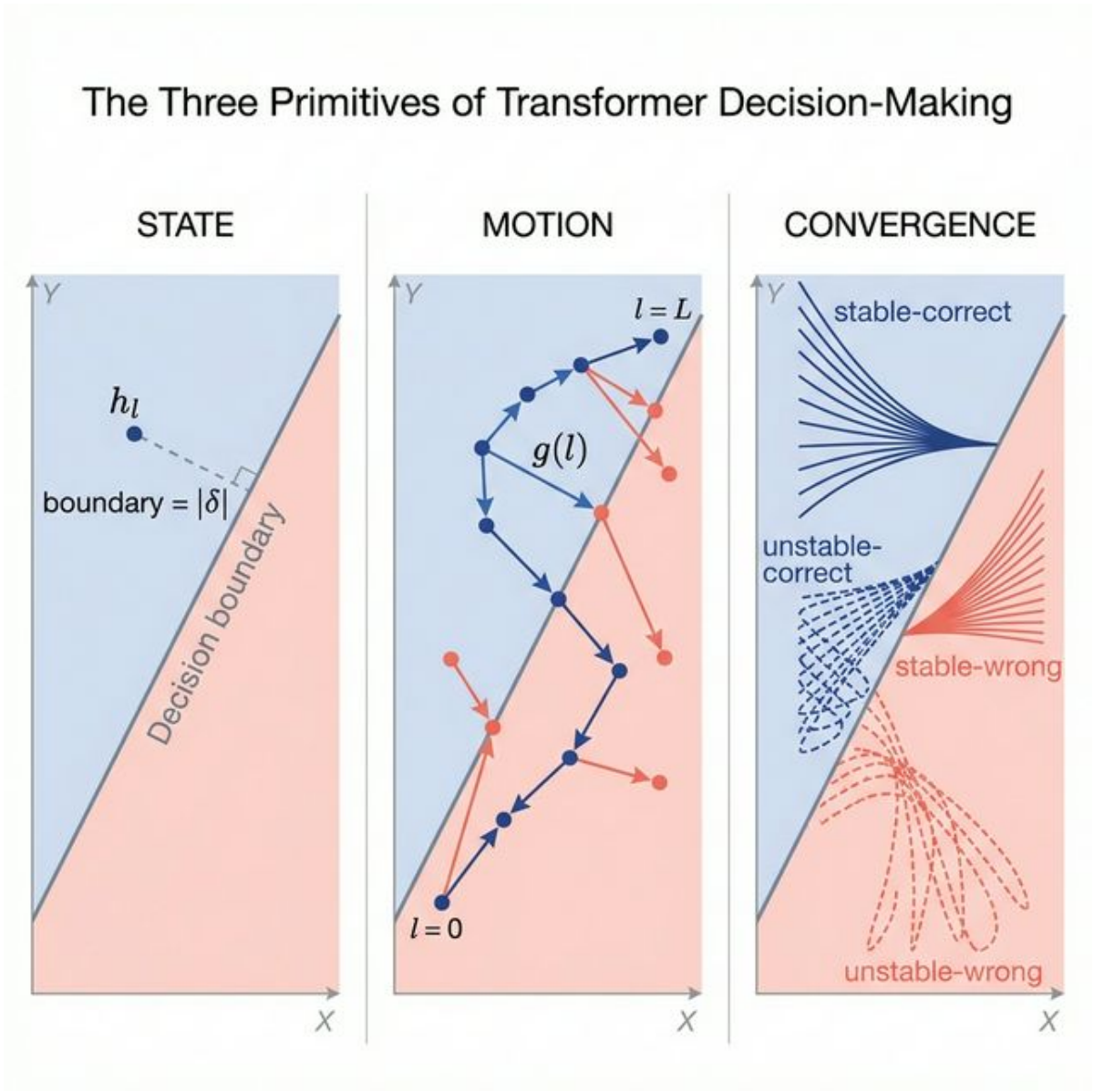

Figure 1. The three primitives and four trajectory types. Left: The state $\delta(l)$ is a point in logit-margin space; the boundary $|\delta|$ measures distance to a decision flip. Center: Motion $g(l)$ describes layer-by-layer drift toward or away from the correct answer. Right: Four trajectory types arise from combinations of final correctness and tail stability.

3. Experimental Setting

3.1. Models

We study three instruction-tuned transformers spanning distinct pre-training recipes at controlled scale (Table 1). All are loaded at pinned HuggingFace revisions with deterministic decoding (temperature = 0, do_sample=false, seed = 12345). Left-padding with explicit position IDs is enforced for batched inference.

Table 1. Models used in this study, with pinned revisions.

Model	Params	Layers
Qwen 2.5-7B-Instruct	7.6B	28
Llama 3.1-8B-Instruct	8.0B	32
Mistral 7B-v0.3	7.2B	32

3.2. Data

We draw 3,000 four-choice questions from MMLU (Hendrycks et al., 2021), balanced across difficulty levels and knowledge domains. The manifest is SHA-256 hashed and version-locked before inference.

3.3. Logged Quantities

For each prompt × model pair we record layerwise logits, probabilities, and entropy for the four candidate tokens at every transformer layer (embedding layers excluded).

For a tracing subset (600 prompts per model, balanced over trajectory type and difficulty), we additionally capture per-layer attention outputs, MLP outputs, and attention weight matrices.

3.4. Why Logit Space?

δ, g, and b are computed directly from logged logits; trajectory type is a deterministic function of these. The attention/MLP decomposition involves a linear fit, and causal experiments use counterfactual simulation. We maintain this distinction throughout.

Softmax probabilities compress high-confidence decisions into a near-saturated regime. Logit margins preserve ordinality and scale. Probability-space analyses appear in Appendix B.

4. Phase Diagram of Convergence

4.1. Four Trajectory Outcomes

Each prompt × model pair is classified into exactly one of four types (Table 2).

Table 2. The four trajectory types.

Type	Corr.	Stab.	Interpretation
Stable-corr.	✓	✓	Knew, stayed confident
Stable-wrong	×	✓	Committed to wrong ans.
Unstable-corr.	✓	×	Oscillated, landed right
Unstable-wrong	×	×	Oscillated and failed

4.2. Phase Structure in $(\delta, \Sigma g)$ Space

Plotting the final-layer margin $\delta(L)$ against cumulative drift Σg over the last 8 layers produces four well-separated clusters (Figure 2). Stable-correct trajectories occupy the upper-right quadrant (large positive margin, reinforcing drift); stable-wrong occupy the lower-left. Unstable types are sparser, near the axes.

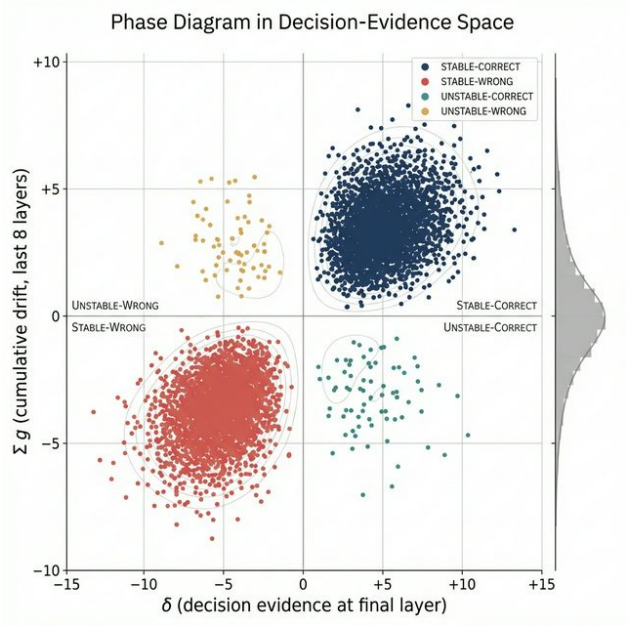

Figure 2. Phase diagram in decision-evidence space. Each point is one prompt × model pair. The four types separate cleanly in the (δ, Σg) plane.

4.3. Cross-Model and Cross-Domain Replication

The phase structure replicates qualitatively across all three models. Absolute magnitudes differ—Qwen produces larger $|\delta|$ than Mistral—but the four-quadrant geometry and bimodal Σg marginal are consistent. Domain shifts the proportion of types but not the geometry.

Caveat. The clean separation partly reflects our threshold choices. Clusters remain visually distinct under perturbations ($\tau \in \{6, 8, 10\}$, $b_{\min} \in \{0.1, 0.3, 0.5\}$), but the boundary between stable and

unstable is a convention, not a phase transition.

5. Opening the Box: Two-Force Decomposition

5.1. Setup

At each layer, the residual stream update sums two components: the attention sub-layer output and the MLP sub-layer output. We extract both at the answer-token position and project each onto a decision direction:

$$d = e_c - e_{k^*}, \tag{4}$$

where e_c and e_{k^*} are one-hot vectors for the correct and competitor tokens. The attention scalar $s_{\text{attn}}(l)$ and MLP scalar $s_{\text{mlp}}(l)$ are the dot products of the respective outputs with d after applying the unembedding matrix.

5.2. Drift Reconstruction

Observed drift should be approximated by:

$$g(l) \approx \alpha \cdot s_{\text{attn}}(l) + \beta \cdot s_{\text{mlp}}(l) + \varepsilon. \tag{5}$$

We fit via OLS on a 70/30 train/test split (at the prompt level). The pipeline requires $R^2 \geq 0.70$ on test data; this gate was pre-registered.

Why linear? A richer model would fit better but explain less. The residual ε captures layer-norm effects, cross-term dependencies, and nonlinear interactions.

5.3. Which Component Drives Which Outcome?

Figure 3 reveals two key patterns:

Attention dominates early steering. In layers 0–15, the attention scalar accounts for the majority of positive drift in stable-correct trajectories—consistent with "routing" evidence-bearing spans toward the answer position.

MLP dominates late commitment. In layers 20–28, the MLP scalar grows in magnitude and can drive the final push toward or away from the correct answer. For stable-wrong trajectories, MLP injection maintains wrong-answer commitment in late layers.

For unstable trajectories, the two components are often misaligned: attention pushes one direction while MLP pushes the other, producing oscillatory drift.

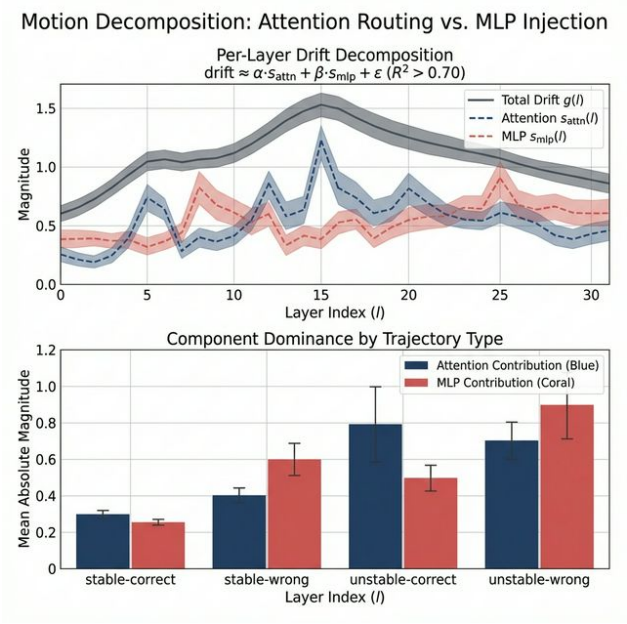

Figure 3. Top: Per-layer drift decomposition into attention and MLP components with confidence bands ($R^2 > 0.70$). Bottom: Mean absolute component magnitudes by trajectory type.

6. Attention as Routing

6.1. Prompt Span Taxonomy

Multiple-choice prompts have natural structure: instruction, question stem, and option spans (A–D). We parse these via regex-based boundary detection.

6.2. Span Labeling via Counterfactual Effect

Rather than hand-labeling, we define spans operationally. For each span s, we delete it and re-run inference. The span effect is $\text{effect}(s) = \delta_{\text{full}} - \delta_{\text{mutated}}$. Spans with effect ≥ 0.05 are labeled evidence; ≤ -0.05 are distractor; otherwise neutral.

Label stability under paraphrase requires $\geq 85\%$ agreement and ≥ 0.80 Jaccard similarity.

6.3. Attention Mass vs. Contribution

We distinguish attention mass (raw weight from answer position to span tokens) from decision-aligned contribution (the attention output projected onto d).

Figure 4 shows that mass is necessary but not sufficient. Stable-correct trajectories route mass to the correct-option span with large positive contribution. Stable-wrong trajectories route to the wrong-option span.

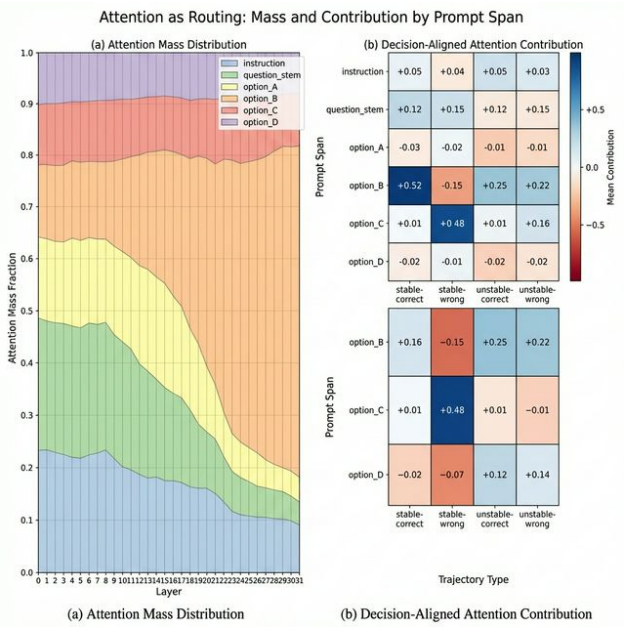

Figure 4. Left: Attention mass by span across layers. Right: Decision-aligned contribution heatmap by trajectory type.

7. MLP as Injection

The MLP scalar $s_{\text{mlp}}(l)$ measures how much parametric knowledge each layer's feedforward network contributes to the decision margin. Unlike attention, MLP can introduce information absent from the input—stored associations, factual recall, and learned biases.

Help regime (layers 5–20, stable-correct): MLP scalars are moderately positive, reinforcing what attention has routed.

Harm regime (layers 20–28, stable-wrong): MLP scalars are strongly aligned with the wrong answer, actively opposing correct-option routing. This is pronounced in STEM domains where the MLP appears to inject confidently wrong associations.

Garbage in, amplified out. For unstable-wrong trajectories, attention routes noisy signals in early layers; MLP amplifies this noise in later layers. Instability is created, not merely revealed, by MLP acting on a poorly conditioned routing signal.

8. Causal Validation

Sections 5–7 are correlational. We conduct three families of causal experiments.

8.1. Component Ablation

We simulate ablating a component by subtracting its scalar from drift at target layers ($l = 20, \ldots, 27$) and recomputing the counterfactual final margin:

$$\delta_L^{\text{cf}} = \delta_0 + \sum_{l=0}^{L-1} [g(l) - \mathbb{1}[l \in T] \cdot s_{\text{comp}}(l)]. \qquad (6)$$

Both attention and MLP ablation produce finite, directionally consistent shifts across all three models (Figure 5a).

8.2. Activation Patching

We patch the component scalar from a successful trajectory into a failing trajectory at target layers. Patching attention from stable-correct into stable-wrong produces positive $\Delta\delta$ for the majority of pairs (Figure 5b). MLP patching is less consistent, reflecting prompt-specific knowledge.

8.3. Span Deletion Tests

Evidence-labeled spans produce significantly larger effects than distractor-labeled spans (Figure 5c). The gap remains positive after Benjamini–Hochberg correction ($\alpha = 0.05$); permutation-test $p < 0.01$ across all three models.

8.4. Negative Controls

Two controls: (1) shuffled—permuting span labels 256 times; (2) sign-flipped—randomly flipping effect signs 256 times. Both baselines center near zero. The observed evidence mean exceeds both by ≥ 0.05, a preregistered margin.

Caveat. Our experiments use counterfactual simulation—not live intervention. Simulated ablation assumes sufficient linearity. Live ablation would be stronger but introduces nonlinear propagation effects.

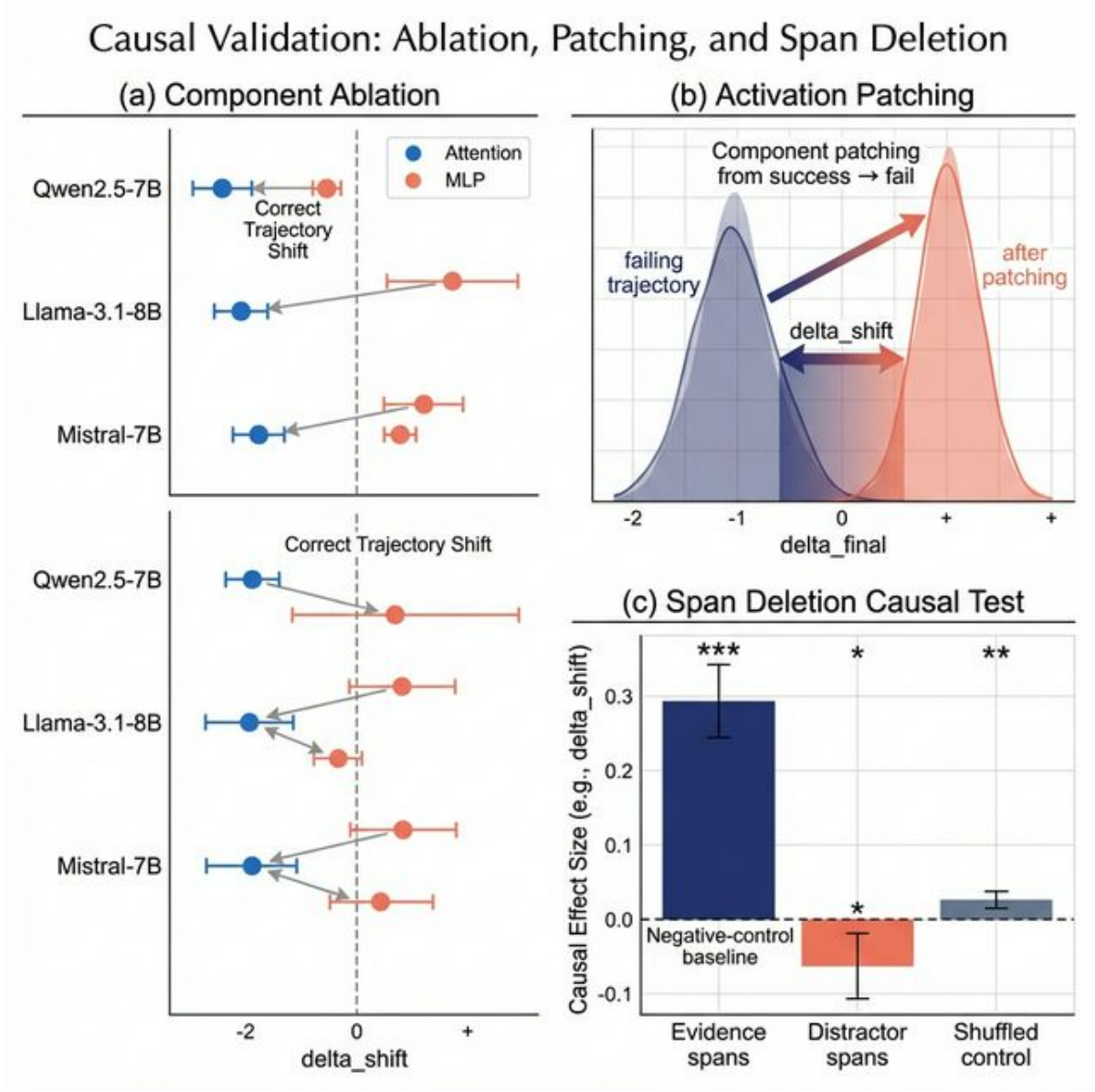

Figure 5. Causal validation. (a) Component ablation: ablating attention from stable-correct decreases δ; ablating MLP from stable-wrong increases δ. (b) Activation patching shifts failing trajectories positive. (c) Evidence spans produce larger causal effects than distractor or shuffled controls ($^{***}p < 0.001$).

9. Discussion and Limitations

What the primitives buy. Rather than "the model got it wrong," we can say "the model entered the correct region at layer 12 but was pushed out by MLP injection at layer 23." This localizes failure, identifies the responsible component, and suggests targeted interventions.

Implications for training. We refrain from strong claims but note a hypothesis: training procedures that penalize drift disagreement between attention and MLP in late layers might reduce stable-wrong rates.

Limitations.

1. Logit-space projection. All quantities live in a 4D logit subspace; full hidden-state structure may be missed.
2. Linear decomposition. Up to 30% of variance remains in ε.
3. Simulated interventions. We simulate rather than execute ablations.
4. Scale. Only 7–8B models studied.
5. Prompt format. Only four-choice MC; open-ended generation may differ.
6. Threshold sensitivity. Classification depends on pre-registered but conventional thresholds.
7. Single run. Results from one deterministic run (seed 12345).

10. Related Work

Logit lens and tuned lens. Nostalgebraist (2020) and Belrose et al. (2023) project intermediate representations through the unembedding matrix. We analyze the trajectory of projections across layers in the answer-token subspace.

Mechanistic interpretability. The circuits paradigm (Elhage et al., 2021; Olsson et al., 2022; Conmy et al., 2023) identifies computational subgraphs. We characterize trajectory-level dynamics, trading microscopic precision for macroscopic coverage.

Probing classifiers. Linear probes (Belinkov, 2022; Hewitt & Manning, 2019) test whether representations encode specific properties. Our $\delta(l)$ is a targeted probe without learned-classifier ambiguity.

Causal tracing. Meng et al. (2022) localize factual associations. Our span-deletion tests operate at the prompt level; ablations target component types rather than individual positions.

Decision dynamics. Recent work (Din et al., 2023; Geva et al., 2023) studies layerwise prediction evolution. We contribute a unified framework connecting descriptive dynamics, mechanistic decomposition, and causal validation.

11. Conclusion

A transformer's decision on a multiple-choice question is not a point but a trajectory. Three measurable quantities—the logit margin δ, its per-layer drift g, and the boundary distance $|\delta|$—suffice to classify this trajectory into four outcome types, producing a phase diagram that replicates across architectures and domains. Decomposing drift into attention routing and MLP injection reveals that attention steers early, MLP commits late, and their disagreement creates instability. Causal validation confirms directionally consistent effects.

The framework is deliberately modest: we claim sufficiency for classification, not necessity; causal relevance, not sole causation; and insight from logit-margin space, not complete representation coverage. The phase diagram is a map. The two-force decomposition is a compass. Together, they provide a vocabulary for navigating the space between a prompt and an answer.

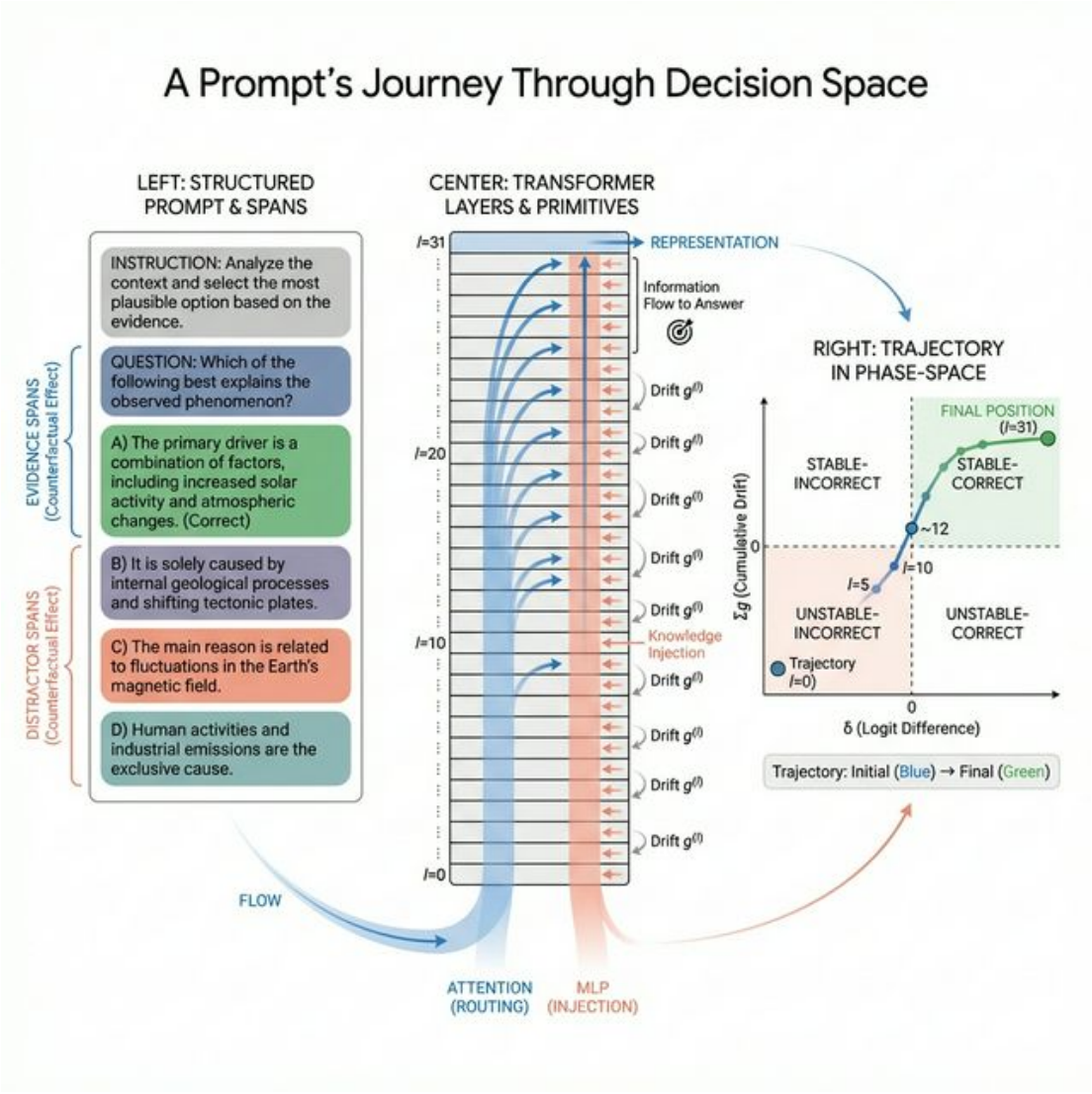

Figure 6. A prompt's journey through decision space. Left: Structured input with evidence and distractor spans identified via counterfactual deletion. Center: Layer-by-layer attention routing and MLP injection. Right: Convergence trajectory in the (δ, Σg) phase diagram.

Impact Statement

This paper presents work whose goal is to advance the field of mechanistic interpretability by providing diagnostic tools for understanding transformer decision-making. Improved understanding of model failures may contribute to safer and more reliable AI systems. We do not foresee specific negative societal consequences beyond those generally associated with advances in machine learning.

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A. Threshold Sensitivity Analysis

We vary the trajectory classification parameters and verify that the four-cluster structure persists.

Tail length $\tau \in \{6, 8, 10\}$: Shorter tails classify more trajectories as stable; longer tails classify more as unstable. Phase geometry is preserved.

Minimum boundary $b_{\min} \in \{0.1, 0.3, 0.5\}$: Lower thresholds admit near-boundary trajectories as stable; higher thresholds reclassify them. Separation degrades gracefully.

Late flip threshold $\in \{0, 1\}$: Allowing one flip expands the stable class modestly for $\tau = 8$.

B. Probability-Space Comparison

Replicating the phase diagram using probability margins $(p_c - p_{k^*})$ in place of logit margins produces visible but less separated clusters. High-confidence decisions compress into a narrow probability range, reducing discrimination. We recommend logit-space analysis for trajectory work.

C. PCA Visualizations of Hidden States

PCA on full hidden-state vectors at selected layers shows partially separable trajectory-type clusters in PC1–PC2 space. However, explained variance is below 15% for PC1, making these visualizations illustrative rather than evidentiary.

D. Decomposition Residual Analysis

The residual ε from Equation (5) has mean near zero with modest layer-dependent heteroscedasticity (larger variance at layers 12–18). No systematic bias invalidates the linear approximation.

E. Per-Domain Breakdowns

STEM domains show higher stable-wrong rates (confident factual errors). Humanities show higher instability (ambiguous or context-dependent questions). Social sciences are intermediate.

F. Methods

Metric computation. For each prompt $\times$ model pair, layerwise candidate logits (A/B/C/D) are extracted from every transformer block. $\delta(l)$, $g(l)$, $b(l)$, and $k^*(l)$ are computed per Section 2. Probabilities are softmax-normalized over four candidates; entropy is $H(l) = -\sum_c p_c(l) \log p_c(l)$ with clipping at 10^{-12}. Correctness derives from the final-layer top-1 logit.

Trajectory classification. Sign flips are computed on the sign sequence of δ, ignoring zeros. Late flips are restricted to the tail window. Stability requires zero late flips and $\min_{l \in \text{tail}} b(l) \geq 0.3$.

Span parsing. Prompt spans are identified via regular-expression boundary detection, recognizing parenthesized, bracketed, and colon-delimited option markers. Span effects are computed by deletion and re-inference. Labels use symmetric thresholds at ± 0.05. Paraphrase stability is validated on 50 prompts per model.

Tracing and decomposition. For the tracing subset, attention and MLP outputs are captured via forward hooks using eager attention. The decision direction d is computed per Equation (4). OLS fit uses a 70/30 train/test split at the prompt level with $R^2 \geq 0.70$ enforced.

Causal experiments. Ablation subtracts component scalars at layers 20–27 per Equation (6). Patching substitutes the failing trajectory's component with the successful trajectory's. Span deletion removes labeled spans and re-measures δ. Negative controls: label-shuffle (256 permutations) and sign-flip (256 iterations).

Statistics. Bootstrap CIs use 2,000 resamples. Permutation tests use 5,000 permutations. Multiple-comparison correction uses Benjamini–Hochberg with $\alpha = 0.05$. All random operations use deterministic seeds.

Reproducibility. The full pipeline runs from a single orchestrator with fail-closed gating. Intermediate artifacts are persisted as Parquet files with SHA-256 manifests. Thermal management enforces checkpoint/resume determinism.